Obstructive vs Restrictive Lung Diseases

Page 1: Spirometry Diagnostics & Flow-Volume Loops

FEV1, FVC, and Diffusion Capacity (DLCO) Differentiation

High-Yield Clinical Concepts & Board Review Points:

- **Obstructive lung defect: FEV1/FVC ratio <0.70 post-bronchodilator. Residual volume (RV) and TLC are increased.**
- **Restrictive lung defect: FEV1/FVC normal or increased (>0.70), but Total Lung Capacity (TLC) is <80% of predicted**
- **DLCO is decreased in Emphysema due to alveolar capillary wall destruction, but NORMAL in chronic bronchitis an**
- **In restrictive disease, DLCO is decreased in intrinsic parenchymal fibrosis (IPF), but NORMAL in chest wall/neurom**

CLINICAL PEARL & EXAM TRAP:

A bronchodilator response is positive if FEV1 or FVC increases by >=12% AND >=200 mL following inhaled albuterol.

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Page 2: COPD Exacerbation Management & Oxygen Targets

Antibiotics, Systemic Corticosteroids, and Non-Invasive Ventilation

High-Yield Clinical Concepts & Board Review Points:

- **Controlled oxygen therapy in acute COPD exacerbations targets an SpO2 of 88-92% to avoid hypercapnic respirato**
- **Non-invasive positive pressure ventilation (BiPAP) is first-line for acute hypercapnic respiratory failure (pH <7.35, P**
- **Five-day systemic corticosteroid course (Prednisone 40 mg daily) accelerates recovery and reduces treatment failu**
- **Antibiotics are indicated when patients have at least 2 cardinal symptoms (increased sputum volume, increased pu**

CLINICAL PEARL & EXAM TRAP:

Macrolides (Azithromycin) or Doxycycline are preferred first-line outpatient oral agents for mild-to-moderate COPD exace